

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. Examination (CE/IT)

November 2013

IT402 WIRELESS COMMUNICATION AND MOBILE COMPUTING

Date: 16.11.2013, Saturday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

Q-1 Answer the following questions.

- [A.] What is super high frequency and extremely high frequency? [1]
[B.] Define the term: Path loss of Radio Signal [1]
[C.] _____ is an effect caused by multipath propagation. [1]
(a) Delay Spread (b) Spectrum Distribution (c) path improvement (d) Sectorizing
[D.] What is use of TMSI and MSRN numbers? [1]
[E.] In GSM, to avoid _____, different transmitters within each other's range uses FDM. [1]
(a) scattering (b) path loss (c) interference (d) crosstalk
[F.] Write down the supplementary services offered by GSM. [1]
[G.] What are the application areas of multiplexing and modulation? [1]
[H.] What are the two basic reasons for handover? [1]
[I.] What do you mean by Doppler effect? [1]
[J.] _____ and _____ are the frequency band used in GSM telecommunication network. [1]
[K.] _____ Protocol is used to achieve tunneling in GPRS. [1]
(a) IP-in-IP encapsulation (b) GP-IN-GP encapsulation
(c) GTP (d) GPRS Tun

Q-2 Answer the following questions.

- [A.] Security deals with Authentication, Access Control and Authorization. Explain authentication mechanism used in GSM. [3]
[B.] Explain active v/s passive hotspot and define beacon message. [4]
[C.] What are the advantages and disadvantages of Cellular System (GSM)? [5]

OR

- Q-2 [A.] Which are the different ranges for signal propagation in wireless communication? [3]
[B.] What are the sequence of steps followed by Mobile Originated Call and Mobile Terminated Call? [4]
[C.] Explain functionality of layer of GSM Protocol Stack. [5]

Q-3 Answer the following questions.(Attempt any Two)

- [A.] What do you mean by multiplexing? What are the types of it? What are the advantages of each type of multiplexing? [12]
[B.] Draw architecture of GSM? Explain major functionality of component of GSM architecture.
[C.] Which are the effects of signal propagation in wireless communication?

SECTION-II

Q-4 Answer the following questions.

- [A] What is meaning of Sector Handover and Radio Handover? [1]
- [B] _____ number of active addresses are possible in Bluetooth?
(a) 7 (b) 8 (c) 15 (d) 16 [1]
- [C] Draw only Frame format of Minimal IP-in-IP encapsulation. [1]
- [D] Which one of the following is true for Least Interference Routing for Adhoc network? [1]
 - (a) It forces to change TCP to find shortest Path
 - (b) It uses node bandwidth to find shortest Path
 - (c) It uses node interference to find shortest path
 - (d) It uses node references to find shortest path
- [E] In Mobile IP, Co-located Care of Address, _____ acts as a tunnel end point. [1]
 - (a) Foreign Agent (b) Home Agent (c) Mobile Agent (d) Correspondent Agent
- [F] Which one of the following is feature of Transaction Oriented TCP? [1]
 - (a) Retransmits Only Lost Data (b) Avoids slow-start after roaming
 - (c) Combines connection (d) Splits TCP Connectionssetup/release and data transmission
- [G] Write down the two sub-layers of Physical layer in IEEE 802.11. [1]
- [H] Which one is most disadvantage of Indirect TCP? [1]
- [I] Why do you say that "Transparency" is requirement to develop Mobile IP concept? [1]
- [J] Write down full form: BSS, ESS [1]
- [K] In Bluetooth, _____ provides three different types of logical channels (connectionless, connection oriented, signaling) that are transported via the ACL between master and slave. [1]
 - (a) L2CAP (b) LMP (c) SDP (d) CRP

Q-5 Answer the following questions.

- [A] Which are the group of functions covered by Link Manager Protocol In Bluetooth? [3]
- [B] What are the advantages of HiperLAN2 over IEEE 802.11 Wireless LAN? [4]
- [C] Draw and explain WAP 2.0 architecture. [5]

OR

- Q-5**
- [A] Define: Hidden station problem [3]
 - [B] What do you mean by DHCP Relay? [4]
 - [C] Three Properties of Traditional TCP i.e. Congestion control, Slow Start and Fast retransmit/Recovery cannot be applied to Wireless Network. Why? [3]

Q-6 Answer the following questions.(Attempt any Two)

- [A] Write down the advantages and disadvantages of Snooped TCP. [12]
- [B] When can you apply Dynamic Source Routing? By taking suitable diagram, explain functionality of DSR.
- [C] What do you mean by Tunneling in Mobile IP? For Tunneling, Is registration mandatory? If YES, then write down mechanism of Registration, if NO, Why is it not mandatory?